28 Supersymmetric Versions of the Standard Model

Physical phenomena at energies accessible in today's accelerator laboratories are accurately described by the standard model, the renormalizable theory of quarks, leptons, and gauge bosons, governed by the gauge group $SU(3) \cdot SU(2) \cdot U(1)$, described in Sections 18.7 and 21.3. The standard model is today usually [1] understood as a low-energy approximation to some as-yet-unknown fundamental theory in which gravitation appears unified with the strong and electroweak forces at an energy somewhere in the range of 10^{16} to 10^{18} GeV. This raises the *hierarchy problem*: what accounts for the enormous ratio of this fundamental energy scale and the energy scale ≈ 300 GeV that characterizes the standard model?

The strongest theoretical motivation for supersymmetry is that it offers a hope of solving the hierarchy problem. Quarks, leptons, and gauge bosons are required by the $SU(3) \cdot SU(2) \cdot U(1)$ gauge symmetry to appear with zero masses in the Lagrangian of the standard model, so that the physical masses of these particles are proportional to the electroweak breaking scale, which in turn is proportional to the mass of the scalar fields responsible for the electroweak symmetry breakdown. The crux of the hierarchy problem [1a] is that the scalar fields, unlike the fermion and gauge boson fields, are not protected from acquiring large bare masses by any symmetry of the standard model, so it is difficult to see why their masses, and hence all other masses, are not in the neighborhood of 10^{16} to 10^{18} GeV.

It has been hoped that this problem could be solved by embedding the standard model in a supersymmetric theory. If the scalar fields appear in supermultiplets along with fermions in a chiral representation of some gauge group, then supersymmetry would require vanishing bare masses for the scalars as well as the fermions. All the masses of the standard model would then be tied to the energy scale at which supersymmetry is broken. The hope of a solution of the hierarchy problem along these lines has been the single strongest motivation for trying to incorporate supersymmetry in a realistic theory.

Unfortunately, none of the new particles required by supersymmetric theories have been detected, and no entirely satisfactory supersymmetric version of the standard model has emerged so far. This chapter will describe the attempts that have been made in this direction.

28.1 Superfields, Anomalies, and Conservation Laws

In this section we shall attempt to decide at least tentatively what ingredients should appear in a supersymmetric version of the standard model.

None of the quark and lepton fields of the standard model belong to the adjoint representation of the $SU(3) \cdot SU(2) \cdot U(1)$ gauge group, so they cannot be the superpartners of known gauge bosons and must therefore be included in chiral scalar superfields. We will define U_i, D_i, $\bar{U}_i$, $\bar{D}_i$, N_i, E_i, and $\bar{E}_i$ as the left-chiral superfields whose ψ_L components are the left-handed fields of the quarks of charge $2e/3$ and $-e/3$, of the antiquarks of charge $-2e/3$ and $+e/3$, of the leptons of charge 0 and $-e$, and of the antileptons of charge $+e$, respectively, with i a generation label running over values 1, 2, and 3. (For instance, the spinor components of U_1, U_2, and U_3 are the left-handed fields of the u, c, and

t quarks, respectively.) Of these superfields, U_i and D_i form $SU(2)$ doublets, N_i and E_i also form $SU(2)$ doublets, and the others are $SU(2)$ singlets. The quark superfields form $SU(3)$ triplets and the antiquark superfields form $SU(3)$ antitriplets, with color indices suppressed, and the leptons and antilepton superfields are $SU(3)$ singlets. As mentioned earlier, the particles described by the scalar components of these superfields are known as squarks, antisquarks, sleptons, and antisleptons. There are also the gauginos, the spin 1/2 superpartners of the gauge bosons of $SU(3)$, $SU(2)$, and $U(1)$, respectively known as the gluino, wino, and bino.[1]

We must also add some mechanism that produces a spontaneous breakdown of $SU(2) \cdot U(1)$ and gives mass to all the quarks and leptons as well as to the $W^\pm$ and Z^0. The simplest possibility is to suppose the existence of just two more $SU(2)$ doublets of left-chiral superfields:

$$H_1 = \begin{pmatrix} H_1^0 \\ H_1^- \end{pmatrix}, \qquad H_2 = \begin{pmatrix} H_2^+ \\ H_2^0 \end{pmatrix}, \tag{28.1.1}$$

which appear in the Lagrangian density in linear combinations of the $SU(3) \cdot SU(2) \cdot U(1)$-invariant $\mathcal{F}$-terms:

$$\big[(D_i H_1^0 - U_i H_1^-)\bar{D}_j\big]_{\mathcal{F}}, \qquad \big[(E_i H_1^0 - N_i H_1^-)\bar{E}_j\big]_{\mathcal{F}}, \tag{28.1.2}$$

and

$$\big[(D_i H_2^+ - U_i H_2^0)\bar{U}_j\big]_{\mathcal{F}}, \tag{28.1.3}$$

with obvious contractions of color indices. According to Eq. (26.4.11), a non-vanishing expectation value of the scalar component of H_1^0 gives mass to the charged leptons and charge $-e/3$ quarks, while a non-vanishing expectation value of the scalar component of H_2^0 gives mass to the charge $+2e/3$ quarks. These expectation values of course also give mass to the $W^\pm$ and Z^0 vector bosons and, since the H_1 and H_2 are $SU(2)$ doublets, we automatically get the same successful results for these masses as found in Section 21.3. Note that supersymmetry does not allow the complex conjugates of the H_1 and H_2 left-chiral superfields to appear in the superpotential, so a vacuum expectation value of the scalar component of H_1^0 cannot give masses to the charge $+2e/3$ quarks, and a vacuum expectation value of the scalar component of H_2^0 cannot give mass to the charge $-e/3$ quarks or the charged leptons, which is why both H_1 and H_2 are needed to give masses to all the quarks and leptons.

Of course, there might be more than one of the H_1 and/or H_2 doublets. Their numbers are partly constrained by the condition of anomaly cancellation. We saw in Section 22.4 that the gauge symmetries of the non-supersymmetric standard model are anomaly-free,

[1] As discussed in Section 28.3, the energy scale that characterizes supersymmetry breaking is expected to be considerably higher than the ≈ 300 GeV that characterizes the breaking of $SU(2) \cdot U(1)$, so there is a substantial range of energies in which supersymmetry but not $SU(2) \cdot U(1)$ may be considered to be broken. In this range, the gauginos have masses that are governed by $SU(2) \cdot U(1)$ symmetry, so the neutral electroweak gauginos of definite mass are superpartners of the $SU(2)$ triplet W^0 and the $SU(2)$ singlet B, known as the neutral wino and the bino, rather than superpartners of the Z^0 and the photon. When $SU(2) \cdot U(1)$ breaking is taken into account there is a small mixing of the neutral wino and bino.

as they must be for quantum mechanical consistency, but now there are additional spinor fields in the Lagrangian. The gaugino fields don't create any problems, because their left-handed components belong to the adjoint representation of the gauge group, which is real for all gauge groups. The only problem can arise from the higgsinos—the spin 1/2 components of the superfields (H_1^0, H_1^-) and (H_2^+, H_2^0). The spinor components of each (H_1^0, H_1^-) doublet of superfields produces an $SU(2)$-$SU(2)$-$U(1)$ anomaly proportional to $\sum t_3^2 y = (\frac{1}{2}g)^2(\frac{1}{2}g') + (-\frac{1}{2}g)^2(\frac{1}{2}g') = \frac{1}{4}g^2 g'$, while the spinor components of each (H_2^+, H_2^0) doublet of superfields produces an $SU(2)$-$SU(2)$-$U(1)$ anomaly proportional to $\sum t_3^2 y = (\frac{1}{2}g)^2(-\frac{1}{2}g') + (-\frac{1}{2}g)^2(-\frac{1}{2}g') = -\frac{1}{4}g^2 g'$. *The cancellation of anomalies thus requires an equal number of* (H_1^0, H_1^-) *and* (H_2^+, H_2^0) *doublets.* In this case, all anomalies cancel, including the $U(1)^3$ and $U(1)$-graviton-graviton anomalies. The next section will give an argument that there is in fact just one of each type of doublet.

In a theory constructed along these lines, we have to give up one of the attractive features of the non-supersymmetric standard model, that is, that it *automatically* excludes any renormalizable interactions that violate the conservation of baryon or lepton number. There are several renormalizable supersymmetric $SU(3) \cdot SU(2) \cdot U(1)$-invariant $\mathcal{F}$-terms that could be included in the Lagrangian density that would violate baryon and/or lepton number conservation without violating the $SU(3) \cdot SU(2) \cdot U(1)$ gauge symmetries:

$$\left[(D_i N_j - U_i E_j)\bar{D}_k\right]_{\mathcal{F}}, \qquad \left[(E_i N_j - N_i E_j)\bar{E}_k\right]_{\mathcal{F}}, \tag{28.1.4}$$

and also

$$\left[\bar{D}_i \bar{D}_j \bar{U}_k\right]_{\mathcal{F}}, \tag{28.1.5}$$

with the three suppressed color indices in Eq. (28.1.5) understood to be contracted with an antisymmetric ϵ-symbol to give a color singlet. With all of these interactions present, there would be no clever way to assign baryon and lepton numbers to the squarks and sleptons that would avoid an unsuppressed violation of baryon and lepton number conservation. For instance, the exchange of the scalar boson of the $\bar{D}$ superfield between vertices for interactions (28.1.4) and (28.1.5) would lead to the process $u_L d_R u_R \longrightarrow \bar{e}_R$, observed for instance as $p \longrightarrow \pi^0 + e^+$, at a catastrophic rate that is only suppressed by factors of coupling constants. To avoid this, it is necessary to make an independent assumption that would rule out some or all of the interactions (28.1.4)–(28.1.5).

Note that it is not necessary to rule out *all* of the interactions (28.1.4) and (28.1.5). For instance, suppose that we assume only that baryon number is conserved, with conventional baryon number assignments: the U_i and D_i left-chiral superfields are assigned baryon number $+1/3$; the $\bar{U}_i$ and $\bar{D}_i$ are assigned baryon number $-1/3$; and the L_i, E_i, H_1, and H_2 all are assigned baryon number 0. This would allow the interactions (28.1.4) while forbidding the interactions (28.1.5). Despite appearances, the interactions (28.1.4) alone do not violate the conservation of lepton number, provided the scalar components of the superfields are assigned appropriate lepton numbers. This can be done by assigning lepton number 0 to the N_i and E_i superfields, lepton number -1 to the U_i, D_i, $\bar{U}_i$, and $\bar{D}_i$ superfields, lepton number -2 to the $\bar{E}_i$ superfields, lepton number 0 to the H_1 and H_2 superfields, and lepton numbers -1 and $+1$ to θ_L and θ_R, respectively. (Recall that such symmetries, under which θ transforms non-trivially, are known as R-symmetries.) Then all quarks and leptons have conventional lepton numbers: the fermion components ν_{iL} and e_{iL}, which are the coefficients of θ_L in N_i and E_i, have the lepton numbers $0+1=+1$; the fermion components $\bar{e}_{iR}$ of the $\bar{E}_i$ superfields have lepton numbers $-2+1=-1$, and the quarks and antiquarks have lepton numbers $-1+1=0$. The higgsinos (the fermion components of H_1 and H_2) have lepton numbers $0+1=+1$. On the other hand, the scalar components of the superfields have the same lepton numbers as the superfields themselves, which are unconventional. Further, the $\mathcal{F}$-term of a left-chiral superfield is the coefficient of θ_L^2, so the interactions in Eq. (28.1.4) have lepton numbers $-1+0-1+2=0$ and $0+0-2+2=0$; the H_1 interactions (28.1.2) have lepton number $-1+0-1+2=0$ and $0+0-2+2=0$, respectively; and the H_2 interaction (28.1.3) has lepton number $-1+0-1+2=0$; so none of these interactions violate the conservation of lepton number. Also, the scalar components of H_1 and H_2 have lepton number 0, so their vacuum expectation values also do not violate the conservation of lepton number. With this assignment of lepton numbers, lepton number conservation rules out any renormalizable interactions that would violate baryon number conservation: the interaction (28.1.5) has lepton number $-1-1-1+2=-1$, and so is forbidden.

The interactions (28.1.4) would allow an alternative mechanism for breaking $SU(2)\cdot U(1)$ and giving mass to the charged leptons and charge $-e/3$ quarks: the scalar components of the neutrino superfields N_i might have non-vanishing vacuum expectation values. (With the lepton number assignments of the previous paragraph this expectation value would not violate lepton number conservation, because these scalar components have the lepton number of the N_i superfields, which is zero.) But we cannot rely on this mechanism to let us do without the H_1 superfields altogether, because we still need the H_2 interactions (28.1.3) to give mass to the charge $+2e/3$ quarks and, as we have seen, the cancellation of anomalies requires equal numbers of H_1 and H_2 superfields.

It is usually assumed instead that some symmetries forbid both interactions (28.1.4) and (28.1.5). Obviously, these symmetries could be baryon and lepton number conservation, with conventional assignments of these numbers: U_i, D_i having baryon number $B=1/3$ and lepton number $L=0$, $\bar{U}_i$ and $\bar{D}_i$ having baryon number $B=-1/3$ and lepton number 0, N_i and E_i having lepton number $L=+1$ and baryon number 0, $\bar{E}_i$ having lepton number -1 and baryon number 0, and H_1^0, H_1^- and H_2^+, H_2^0 and θ_L and θ_R all having baryon and lepton numbers 0. The same results apply if we only assume the conservation of certain linear combinations of baryon and lepton number, such as the anomaly-free combination $B-L$ discussed in Section 22.4.

There are widespread doubts about whether it is possible to have exact continuous global symmetries, because in string theory the existence of any exact continuous symmetry would imply the existence of a massless spin 1 particle coupled to the symmetry current, so that the symmetry would have to be local, not global [1b]. But the interactions (28.1.4) and (28.1.5) may also be banned by assuming a *discrete* global symmetry, known as the conservation of R parity [2]. The R parity is defined to be +1 for quarks, leptons, gauge bosons, and Higgs scalars, and -1 for their superpartners. This R parity is equal to

$$\Pi_R = (-1)^F (-1)^{3(B-L)}\,, \tag{28.1.6}$$

where $(-1)^F$ is the fermion parity, which is $+1$ for all bosons and -1 for all fermions. Fermion parity is the same sign as is produced by a 2π rotation, and is therefore always conserved, so if $B-L$ is conserved then so is R parity.[2] It is possible that R parity may be conserved even if $B-L$ is not, but in fact the interactions (28.1.4) and (28.1.5) are forbidden by R parity conservation, so as far as renormalizable interactions are concerned R parity conservation implies the conservation of both baryon and lepton number. This is *not* true of the non-renormalizable supersymmetric interactions that are presumably produced by physical processes at very high energies. The baryon- and lepton-non-conserving processes produced by such interactions are discussed in Section 28.7.

All of the new "sparticles" (squarks, sleptons, gauginos, and higgsinos) that are required by supersymmetry theories have negative R parity, so if R parity is exact and unbroken then the lightest of the new particles required by supersymmetry must be absolutely stable. All of the other new particles will then undergo a chain of decays, ultimately yielding ordinary particles and the lightest new particle. Much of the phenomenology of various supersymmetry models is governed by the choice of which of the new particles is the lightest.

With supersymmetry and either R parity or $B-L$ conserved, the most general renormalizable Lagrangian for the superfields discussed above consists of the usual gauge-invariant kinematic part for the chiral superfields, given by a sum of terms of the form $(\Phi^* \exp(-V)\Phi)_D$ for each of the quark, lepton, and Higgs chiral superfields, plus the usual gauge-invariant kinematic term for the gauge superfields, given by a sum of terms of the form $\epsilon_{\alpha\beta}(W_\alpha W_\beta)_{\mathcal{F}}$ for each of the $SU(3)$, $SU(2)$, and $U(1)$ field strength superfields, plus the supersymmetric Yukawa couplings, given by a linear combination of the interactions (28.1.2), (28.1.3), and a new $\mathcal{F}$-term coupling H_1 and H_2:

$$\begin{aligned}\mathcal{L}_Y = &\sum_{ij} h^D_{ij}\left[(D_i H_1^0 - U_i H_1^-)\bar{D}_j\right]_{\mathcal{F}} + \sum_{ij} h^E_{ij}\left[(E_i H_1^0 - N_i H_1^-)\bar{E}_j\right]_{\mathcal{F}} \\ &+ \sum_{ij} h^U_{ij}\left[(D_i H_2^+ - U_i H_2^0)\bar{U}_j\right]_{\mathcal{F}} + \mu\left[H_2^+ H_1^- - H_2^0 H_1^0\right]_{\mathcal{F}} + \text{H.c.}\end{aligned} \tag{28.1.7}$$

As we will see in Section 28.3, more terms will have to be added to the Lagrangian in order to account for supersymmetry breaking.

The coefficient μ in Eq. (28.1.7) has the dimensions of mass, and is the only dimensional parameter that enters in the supersymmetric version of the standard model Lagrangian. It is somewhat disappointing to find that this term is still allowed, because it revives the hierarchy problem: why is μ not of order 10^{16} to 10^{18} GeV? The μ-term in Eq. (28.1.7) can be avoided if we assume that lepton number is conserved, with the unconventional lepton number assignments discussed above that would allow the interactions (28.1.4) but not the

[2] The value of $(-1)^{3(B-L)}$ is -1 for quark and lepton superfields and $+1$ for all other superfields, so the conservation of R parity is equivalent to invariance under a transformation in which all quark and lepton superfields change sign with no change in other superfields. This invariance principle was introduced in Reference 3 in order to rule out the interactions (28.1.4) and (28.1.5).

interactions (28.1.5). In this case, the μ-term carries lepton number $+2$ and is therefore also forbidden. This term can also be forbidden if we assume a $U(1)$ Peccei–Quinn symmetry [4], for which the superfields H_1 and H_2 carry equal quantum numbers, say $+1$, while θ_L and θ_R are neutral. The interactions (28.1.2) and (28.1.3) which give mass to the quarks and leptons are then allowed if, for instance, we give Peccei–Quinn quantum numbers -1 to the left-chiral superfields of antisquarks and antisleptons while the left-chiral superfields of squarks and sleptons are taken to be neutral. This choice then also forbids the dangerous interactions (28.1.4) and (28.1.5). Unfortunately, as we will see in Section 28.4, the μ-term in Eq. (28.1.7) seems to be needed for phenomenological reasons. The theories of gravity-mediated supersymmetry breaking discussed in Section 31.7 provide a natural mechanism for producing a μ-term of an acceptable magnitude.

We can obtain a crude upper bound on the masses of the new particles by assuming that supersymmetry does solve the hierarchy problem discussed at the beginning of this chapter. In accord with the theorem of Section 27.6, if supersymmetry were unbroken the contribution to the mass of the scalar component of H_1 or H_2 from one-loop diagrams with an intermediate quark, lepton, W or Z loop would be cancelled by the corresponding one-loop diagram with an intermediate squark, slepton, wino, or bino. Therefore with supersymmetry broken the contribution δm_H^2 of such diagrams to the squared masses of the H_1 and H_2 scalars is a sum of terms of order $(\mathcal{G}_s^2/8\pi^2)\Delta m_s^2$, where $\mathcal{G}_s$ is the Yukawa or gauge coupling of the Higgs scalar to supermultiplet s, and Δm_s^2 is the mass-squared splitting within the supermultiplet. To avoid having to fine-tune these corrections, we need δm_H^2 not much larger than the coefficient of order $(300\text{ GeV})^2$ of the term in the standard model Lagrangian density that gives the observed $SU(2)\cdot U(1)$ breaking in the tree approximation, so we will assume that $\delta m_H^2 < (1\text{ TeV})^2$. For instance, the top quark and squark have couplings to H_2 of order unity, so we expect that the splittings Δm^2 should be less than about $8\pi^2$ TeV^2, and hence the masses of the top squarks should be less than about 10 TeV. We will see in Section 28.4 that the rates of flavor-changing processes can be brought within experimental upper bounds by taking the masses of the squarks to be nearly equal, in which case this can be taken as a rough upper limit on the masses of all the squarks. (However it is possible that the rates of these processes may be suppressed instead by very large masses of the first two generations of squarks, while the mass of the top squark is below the 10 TeV naturalness bound [4a].) The limits set by this sort of argument on the masses of other particles with $R = -1$ are somewhat weaker, but at least in the popular class of models discussed in Section 28.6, none of the masses of these particles are expected to be much greater than the squark masses, so 10 TeV can be taken as an upper bound on all of them. On the other hand, the fact that none of these particles have been observed only indicates that their masses are probably greater than about 100 GeV, so there is an ample mass range in which they may yet be found.

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If R parity conservation or some other conservation law makes the lightest of the new particles predicted by supersymmetry stable, then some of these particles may be left from the early universe. The number density of these relics can be estimated using techniques that were originally applied to the cosmic density of massive neutrinos [4b]. To give one example of this sort of calculation, we shall show that for a broad range of plausible masses, the new stable particle of supersymmetry theories cannot be a charged and uncolored particle, like a charged slepton, wino, or higgsino [4c].

Once the cosmic temperature T (in energy units, with the Boltzmann constant set equal to unity) drops below the mass m of any stable charged untrapped particle, their number nR^3 in a volume R^3 that expands with the universe is decreased by annihilation at a rate per particle equal to $\overline{v\sigma}n$, where $\overline{v\sigma}$ is the mean value of the product of the relative velocity and the annihilation cross-section. That is,

$$\frac{d(nR^3)}{dt} = -\overline{v\sigma}\, n^2 R^3 \,,$$

so that

$$\frac{1}{nR^3} = \left(\frac{1}{nR^3}\right)_0 + \int_{t_0}^{t} \frac{\overline{v\sigma}}{R^3}\, dt \,, \tag{28.1.8}$$

where 0 labels the epoch at which $T \simeq m$. The annihilation process is exothermic, so $\overline{v\sigma}$ approaches a constant for $v \ll 1$. Also, in a radiation-dominated phase of the cosmic expansion $R \propto t^{1/2}$, so the integral converges, and gives

$$\begin{aligned}\left(\frac{1}{nR^3}\right)_{t\to\infty} &= \left(\frac{1}{nR^3}\right)_0 \overline{v\sigma} \int_{t_0}^{\infty} \frac{dt}{R_0^3 (t/t_0)^{3/2}} \\ &= \left(\frac{1}{nR^3}\right)_0 \frac{2\overline{v\sigma} t_0}{R_0^3} \,.\end{aligned} \tag{28.1.9}$$

The density n_B of baryon number (baryons minus antibaryons) goes as R^{-3}, so this can be rewritten as a formula for the present ratio of new particles to baryons:

$$(n/n_B)_\infty = \left[(n_B/n)_0 + 2\overline{v\sigma}\, n_{B0} t_0\right]^{-1} . \tag{28.1.10}$$

We expect that the ratio $(n/n_B)_0$ at the time that T drops to a value $\approx m$ is roughly of order unity, and since in any realistic theory the present ratio $(n/n_B)_\infty$ must be much less than unity, we can neglect the first term in the denominator on the right-hand side of Eq. (28.1.10), and write instead

$$(n/n_B)_\infty \simeq \frac{1}{\overline{v\sigma}\, n_{B0} t_0} \,. \tag{28.1.11}$$

The precise value of $\overline{v\sigma}$ depends on the particle spin and its interactions; keeping track only of factors of 2π, the particle mass m, and the electric charge, we can estimate it generally to be of order

$$\overline{v\sigma} \approx \frac{e^4 \mathcal{N}}{2\pi m^2} \approx 10^{-3} \frac{\mathcal{N}}{m^2} \,, \tag{28.1.12}$$

where $\mathcal{N}$ is the number of charged particle spin states with mass less than m, into which this particle may annihilate. Also, the age of the universe at a temperature $T_0 \simeq m$ is $t_0 \simeq m_{\rm PL}/m^2$, where $m_{\rm PL} \simeq 10^{18}$ GeV, and the density of baryon number is about 10^{-9}

times the photon number density, which is of order T^3, so that $n_{B0} \simeq 10^{-9} m^3$. Putting this together, we find the present ratio of the new charged particles to baryons:

$$(n/n_B)_\infty \approx 10^{12} \frac{m}{m_{\rm PL}\mathcal{N}} \approx 10^{-6} \frac{m\,({\rm GeV})}{\mathcal{N}}\,. \tag{28.1.13}$$

These new charged particles would experience the same condensations into galaxies, stars, and planets as ordinary baryons, so this would be the ratio observed today on earth. But experiments [4d] that apply mass spectroscopy to samples of water that have been strongly enriched in heavy water-like molecules by electrolysis have set limits of about $10^{-21} n_B$ on the number density of new charged particles with 6 GeV $< m <$ 330 GeV in terrestrial matter. Thus, even if $\mathcal{N}$ is as large as 1000, these measurements decisively rule out the existence of any new charged untrapped particles in this mass range in the numbers that would have been left over from the early universe.

On the other hand, neutral untrapped particles would be left in intergalactic space. Such particles might well provide the "missing mass," that seems to be necessary to account for the gravitational field that governs the motion of galaxies in clusters of galaxies. One of these possible neutral particles is the gravitino, whose cosmological abundance is discussed in Section 28.3. Ellis *et al.* [4c] have extended cosmological considerations to all the new particles required by supersymmetry.